Simplex Method

- A mathematical procedure for solving linear programming problems.
- Simplex method can handle two or more decision variables.
- Simplex Method of Linear Programming was developed by George B. Dantzig of Stanford University.

LP in Standard Maximum Form

(Maximization Problem)

A linear programming problem is in **standard form** if

a. the objective function is to be maximized

b. all decision variables are nonnegative

c. all constraints involve ≤

d. the constants on the right side in the constraints are all positive.

Simplex Method

Steps:

1. Determine the objective function.
2. Write down all necessary constraints.
3. Convert each constraint into an equation by adding a **slack variable.**
4. Set up the initial simplex tableau.
5. Select **pivot column** by finding the **most negative indicator**. (**Indicators** are those elements in bottom row)
6. Select **pivot row**. (Divide the **last column** by **pivot column** for each corresponding entries). Choose the **smallest positive result**. The corresponding row is the **pivot row**. In case there is no positive result, there is **no optimal solutions**. Stop!
7. Find **pivot**. Circle the pivot entry at the **intersection** of the pivot column and the pivot row, and identify **entering variable** and **exit variable** . **Divide pivot** by **itself** in that row to **obtain 1**. Also **obtain zeros for all rest entries in pivot column.**
8. Perform row operation using **Pivoting Method.**

Example: ***(TENT PRODUCTION PROBLEM)***

Maximize : $P = 500X_1 + 800\,X_2$

Subject to: $X_1 + 2X_2 \leq 32$

$3X_1 + 4X_2 \leq 84$

$X_1, X_2 \geq 0$

The New LP Problem:

Maximize: $P = 500X_1 + 800X_2 + 0\mathbf{S_1} + 0\mathbf{S_2}$

$P - 500X_1 - 800X_2 - 0\mathbf{S_1} - 0\mathbf{S_2} = 0$

Subject to: $X_1 + 2X_2 + \mathbf{S_1} = 32$

$3X_1 + 4X_2 + \mathbf{S_2} = 84$

$X_1, X_2, \mathbf{S_1}, \mathbf{S_2} \geq 0$

Initial Tableau.

Maximize: $P = 500X_1 + 800X_2 + 0\mathbf{S_1} + 0\mathbf{S_2}$

$P - 500X_1 - 800X_2 - 0\mathbf{S_1} - 0\mathbf{S_2} = 0$

Subject to: $X_1 + 2X_2 + \mathbf{S_1} = 32$

$3X_1 + 4X_2 + \mathbf{S_2} = 84$

$X_1, X_2, \mathbf{S_1}, \mathbf{S_2} \geq 0$

Basic	P	X_1	X_2	S_1	S_2	RHS
S_1	0	1	2	1	0	32
S_2	0	3	4	0	1	84
P	1	-500	-800	0	0	0

Initial Tableau:

Basic	P	X_1	X_2	S_1	S_2	RHS
S_1	0	1	2	1	0	32
S_2	0	3	4	0	1	84
P	1	-500	-800	0	0	0

Pivot row

Smallest positive Ratio:
32÷ 2 = **16**
84 ÷ 4= 21

Basic variables

Pivot column

Pivot element

Entering variable: X_2

Leaving variable : S_1

Initial solution:

$S_1 = 32$ $X_1 = 0$

$S_2 = 84$ $X_2 = 0$

$P = 0$

First Iteration

Basic	P	X_1	X_2	S_1	S_2	RHS
S_1	0	1	2	1	0	32
S_2	0	3	4	0	1	84
P	1	-500	-800	0	0	0

0-[(-800)(0)÷ 2]

1-[(0)(4)÷ 2]

3-[(4)(1)÷ 2]

-500-[(-800)(1)÷ 2]

0-[(4)(1)÷ 2]

0-[(-800)(1)÷ 2]

84-[(4)(32)÷ 2]

0-[(0)(4)÷ 2]

Basic	P	X_1	X_2	S_1	S_2	RHS
X_2	0	1/2	1	½	0	16
S_2	0	1	0	-2	1	20
P	1	-100	0	400	0	12,800

0-[(-800)(32)÷ 2]

1-[(0)(-800)÷ 2]

Basic	P	X_1	X_2	S_1	S_2	RHS
X_2	0	1/2	1	½	0	16
S_2	0	1	0	-2	1	20
P	1	-100	0	400	0	12,800

Smallest positive ratio:

$16 \div \frac{1}{2} = 32$

$20 \div 1 =$ **20**

Entering variable: $\boldsymbol{X_1}$

Leaving Variable: $\boldsymbol{S_2}$

Second Iteration:

Basic	P	X_1	X_2	S_1	S_2	RHS
X_2	0	0	1	3/2	-1/2	6
X_1	0	1	0	-2	1	20
P	1	0	0	200	100	14,800

Optimal solution

X_2 = 6 **S_1 = 0**

X_1 = 20 **S_2 = 0**

MaxP = 14,800

Solve the following linear programming problem using simplex method.

1. Maximize $P = 100X_1 + 300X_2 + 200X_3$

   Subject to:

$$X_1 + X_2 + X_3 \leq 100$$

$$40X_1 + 20X_2 + 30X_3 \leq 3200$$

$$X_1 + 2X_2 + X_3 \leq 160$$

$$X_1, X_2, X_3 \geq 0$$

New LP problem: $P - 100X_1 - 300X_2 - 200X_3 - 0S_1 - 0S_2 - 0S_3 = 0$

$$X_1 + X_2 + X_3 + S_1 = 100$$

$$40X_1 + 20X_2 + 30X_3 + S_2 = 3200$$

$$X_1 + 2X_2 + X_3 + S_3 = 160$$

$$X_1, X_2, X_3, S_1, S_2, S_3 \geq 0$$

Initial Tableau:

Basic	P	X1	X2	X3	S1	S2	S3	RHS
S1	0	1	1	1	1	0	0	100
S2	0	40	20	30	0	1	0	3200
S3	0	1	2	1	0	0	1	160
P	1	-100	-300	-200	0	0	0	0

Smallest positive ratio:
100 ÷1 = 100
3200 ÷20 = 160
160 ÷2= **80**

Entering variable: X_2
Leaving Variable: S_3

1st Iteration:

Basic	P	X_1	X_2	X_3	S_1	S_2	S_3	RHS
S_1	0	1/2	0	1/2	1	0	-1/2	20
S_2	0	30	0	20	0	1	-10	1600
X_2	0	1/2	1	1/2	0	0	1/2	80
P	1	50	0	-50	0	0	150	24000

Smallest positive ratio:
20 ÷1/2 = **40**
1600 ÷20 = 80
80 ÷ 1/2= 160

Entering variable: X_3
Leaving Variable: S_1

1st Iteration:

Basic	P	X_1	X_2	X_3	S_1	S_2	S_3	RHS
S_1	0	1/2	0	1/2	1	0	-1/2	20
S_2	0	30	0	20	0	1	-10	1600
X_3	0	1/2	1	1/2	0	0	1/2	80
P	1	50	0	-50	0	0	150	24000

2nd Iteration:

Basic	P	X_1	X_2	X_3	S_1	S_2	S_3	RHS
X_3	0	1	0	1	2	0	-1	40
S_2	0	10	0	0	-40	1	10	800
X_2	0	0	1	0	-1	0	1	60
P	1	100	0	0	100	0	100	26,000

Optimal Solution:

$X_3 = 40$ $X_1 = 0$

$S_2 = 800$ $S_1 = 0$

$X_2 = 60$ $S_3 = 0$

Maximum Profit: P26,000

Exercise 1.

$$
\begin{aligned}
\text{Maximize} \quad & z = 2x_1 + 3x_2 + x_3 \\
\text{subject to} \quad & x_1 + x_2 + 4x_3 \leq 100 \\
& x_1 + 2x_2 + x_3 \leq 150 \\
& 3x_1 + 2x_2 + x_3 \leq 320 \\
\text{with} \quad & x_1 \geq 0, x_2 \geq 0, x_3 \geq 0
\end{aligned}
$$

2. A social science professor has received a grant to fund a research project involving voting trends in the Philippines. The budget of the grant includes P32,000 for conducting a door-to-door interviews the day before the election. College students, graduate students and faculty members will be hired to conduct the interviews. Each college student will conduct 18 interviews and be paid P1,000. Each graduate student will conduct 30 interviews and be paid P1,500. Each faculty member will conduct 25 interviews and be paid P2,000. Due to limited transportation facilities, no more than 20 interviewers can be hired. How many undergraduate students, graduate students and faculty members should be hired in order to maximize the number of interviews that will be conducted. What is the maximum number of interviews?